

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Senior AI/ML engineer shipping production agentic AI, LLM document intelligence, retrieval workflows, AI guardrails, data platforms, and fintech tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

- Delivered production agentic AI document intelligence, human-in-the-loop review, and multimodal compliance screening for high risk merchant onboarding.
- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
 - Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.
 - Built a NotebookLM-based technical knowledge hub spanning vendor/web documentation and internal monorepo docs; automated corpus refresh with a Colab pipeline that re-ingests the latest sources to keep retrieval accurate and up to date.
 - Drove internal AI adoption with Notion MCP playbooks, high-signal prompt patterns, and weekly office hours reaching ~15% of the company; standardized PII/confidential-document safeguards and auditability requirements.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

- Built enterprise data pipelines, risk analytics, and data-quality automation for credit trading while scaling secure internal AI developer tooling.
- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
 - Devised ML-driven anomaly detection for credit trading and risk analytics, reducing false positives by 25% and manual review time for risk teams.
 - Developed a resilient Python wrapper for KDB+ with CI-style tests and optimized Q functions, democratizing data access for 5,000+ users.
 - Engineered a cloud-function-driven asset-fee data sink, consolidating 10+ sources into a centralized database, cutting manual maintenance by 90% and providing on-demand access for traders through a Python SDK, REST API, and interactive dashboard.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending app with React, Next.js, TypeScript, MobX, and ethers.js for secure wallet and smart-contract workflows.
- Implemented CI/CD with GitHub Actions, Husky, Cypress, and Jest, halving release cycles, raising test coverage to 90%, and mentoring a front-end intern.
- Optimized front-end performance through code splitting, lazy loading, and asset caching, reducing initial load times by 40% and JavaScript bundle size by 30%.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in competitive ecosystem grant funding.
- Built Python pipelines that normalized a 1M+ Reddit posts/comments dataset into a reusable analytics data product for external users.

Demand Forecasting Extern

Gap Inc. | San Francisco, CA | Jan 2020 - May 2020

- Applied time-series decomposition and bias correction techniques to demand forecasts, boosting accuracy by up to 25% and minimizing surplus inventory costs.
- Owned the data engineering layer by architecting scalable MySQL and Apache Spark pipelines across 5+ years of sales data, improving data processing throughput by 95% and standardizing insights across forecasting teams.

SKILLS

MACHINE LEARNING AND GENERATIVE AI: scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails

DATA ENGINEERING AND ANALYTICS: Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, DuckDB, Polars, Parquet

PROGRAMMING & SCRIPTING LANGUAGES: Python, SQL, TypeScript, JavaScript, Go, Rust, Bash

CLOUD INFRASTRUCTURE AND DEVOPS: AWS, Docker, Kubernetes, Terraform, GitHub Actions, AWS Lambda, Google Cloud Functions, ECS, EKS

PROJECTS

AI Agent Harness Configs | github.com/wyattowalsh/agents

Python/TypeScript MCP developer-workflow catalog with 40+ reusable skills, FastMCP configs, shared instructions, and cross-runtime agent automation.

- Built 40+ instruction and skill packs plus harness configs across Claude Code, Codex, GitHub Copilot, Cursor, and related IDE agents.
- Shipped generated docs, install flows, and portable tool and MCP presets for code review, security, infrastructure, research, and workflow automation.

NBA Basketball Database | github.com/wyattowalsh/nbadb

Python warehouse using Pandas, Polars, SQLAlchemy, DuckDB, and Pandera over 65K+ NBA games, 13M+ events, Parquet exports, docs, and CLI.

- Implemented Pandera-validated NBA API ETL with daily warehouse updates plus DuckDB, SQLite, Parquet, and CSV exports.
- Built docs, schema references, release workflows, and Kaggle distribution for a 180+ table public warehouse spanning 130+ stats.nba.com endpoint classes.

Listentropy | github.com/wyattowalsh/listentropy

TypeScript/React listening-analytics app using Web Workers, Canvas exports, and Vercel to process private Spotify history in-browser.

- Architected a Web Worker pipeline with stage-level progress tracking for full streaming history datasets in-browser.
- Built a Share Studio with dozen-plus card presets, PNG/ZIP export, themes, and strict performance budgets.

ProxyWhirl | github.com/wyattowalsh/proxywhirl

Python/SQLite/PyPI data-collection toolkit with nine rotation strategies, 100+ auto-validated sources, and self-healing proxy health checks.

- Implemented 9 proxy rotation strategies with configurable fallback chains, per-source scoring, and performance-based selection.
- Built self-healing validation for 100+ proxy sources with circuit breakers, auto-ejection, and recovery promotion.

AI/ML Web Feeds | github.com/wyattowalsh/ai-web-feeds

Python source-discovery platform with Pydantic, SQLAlchemy, YAML feeds, Fumadocs docs, Vercel deploys, HTTP/RSS validation, and OPML exports.

- Defined YAML feed schemas with topic and source-type validation across blogs, papers, and newsletters for reliable feed quality.
- Shipped helper scripts, CI checks, scheduled validation jobs, and live docs for reliable community feed review.

MCP-Crawl4AI | github.com/wyattowalsh/mcp-crawl4ai

Python/FastMCP server using Crawl4AI, Playwright, and Docker so AI agents can crawl, extract, and return structured web content over stdio/HTTP.

- Built a FastMCP v3 server with managed headless Chromium, session-aware crawling, and automatic Playwright setup using two runtime dependencies.
- Exposed stdio/HTTP tools, resources, and prompt templates with docs, tests, and release-tagged MCP install flows.

MDXPad | github.com/wyattowalsh/mdxpad

Electron/React/TypeScript macOS editor with CodeMirror, Vercel AI SDK assistance, synchronized MDX preview, and Keychain credentials.

- Built a dual-format Electron editor with CodeMirror 6, synchronized MDX preview, and YAML frontmatter validation.
- Integrated multi-provider AI assistance with token/cost tracking and Keychain-backed secret storage.

Personal Website: w4w.dev | github.com/wyattowalsh/personal-website

Next.js/React/TypeScript personal web app using Tailwind CSS and MDX for durable content, SEO, polished UI, and authoring workflows.

- Maintained the public site across multiple generations, covering responsive landing pages, themes, motion, accessibility, and MDX publishing.
- Built custom MDX publishing with search, metadata, math rendering, feeds, tags, and integrated authoring tools.

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering and Operations Research (IEOR)
Aug 2014 - Dec 2020

High School Diploma

The Hotchkiss School
Aug 2012 - May 2014

AWS Certified Cloud Practitioner

Amazon Web Services (AWS) | Nov 2024

IBM Data Science Professional Certificate

IBM | Feb 2021